Brushfall: Balance & Economy

Design workbook by Umut Eren Demirci. Blue cells are inputs; everything else is calculated or measured.

Sheet	What it contains	Live formulas?
Charge	Hero power after the 30 s match-3 charge for three players (weak, average, strong), and a calculator for the power rules.	Calculator
Climbs	All 24 climbs: guardian, the smallest board that can win, and each player's win rate, paint and stars, with a chart.	No
Restorations	Each quarter of the three paintings: buckets, paint by colour at the chapter's price, and what the sand is made of.	Price per chapter
Economy	Whole chapters per player: climbs played, extra climbs for paint, paint earned, spent and left.	No
Colours	Paint income by colour against what each chapter's painting needs, at the bucket prices you set.	Yes
Meta	Phase 6: the daily painting's boxes, the weekend Grand Climb's tiers by player, and skin prices against the paint each player has left over.	Skin prices
Tuning	Every tuning number, as the build uses it.	No

Source of truth: the game's code (Assets/_Project/Scripts/Brushfall/Core). Measured data comes from the balance runner (Tools/sim), which compiles those same rules outside Unity and plays them with bots at human speed: 300 charges per player, 300 climbs per player per level, and 40 whole journeys per player for the economy. Players: weak (2.4 s a move, random matches, careful but slips 10% of the time on the climb), average (1.7 s, biggest match), strong (1.1 s, plays for power and specials).

The charge

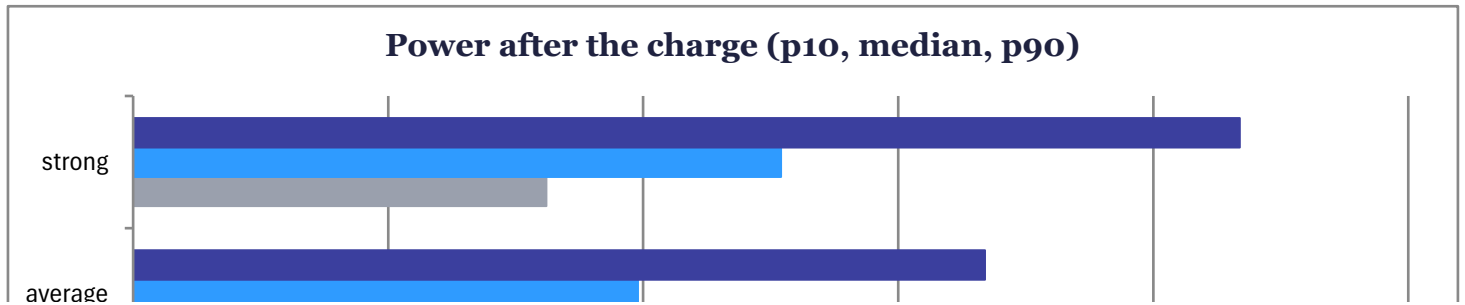
Hero power after 30 seconds of match-3. 300 charges per player.

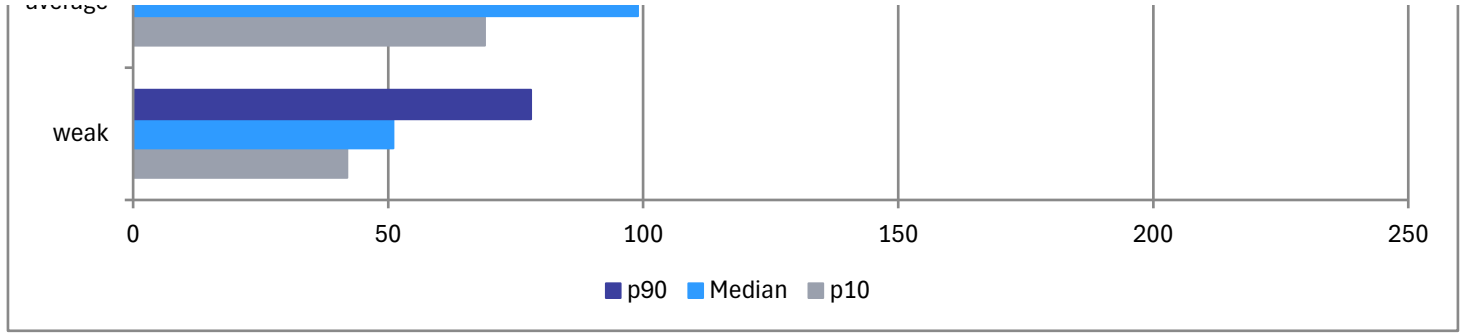
Player	Seconds a move	Power p10	Median	Mean	p90	Starts with a shield	Starts with a magnet	Moves
weak	2.4 s	42	51	57	78	29%	8%	7.2
average	1.7 s	69	99	111	167	66%	21%	7.7
strong	1.1 s	81	127	138	217	83%	42%	9.2

Power calculator

Start power	**20**
Power per tile	**0.5**
Power per special made	**3**
Moves in the charge	**8**
Tiles cleared a move	**5**
Specials made a move	**0.4**
Estimated power	**50**

Phase 5 started with a fast bot charging ~200 power against levels built for 50-140, and shields in 88-100% of runs. At human speed with these rules the medians are about 50 / 100 / 125: the gap between a weak and a strong player is real but each can play every level. Shields now come only from Star specials, so a shield is earned. Combos multiply the whole charge by 1.4, the big moment a strong player chases.




weak
0
50
100
150
200
250
p90
Median
p10

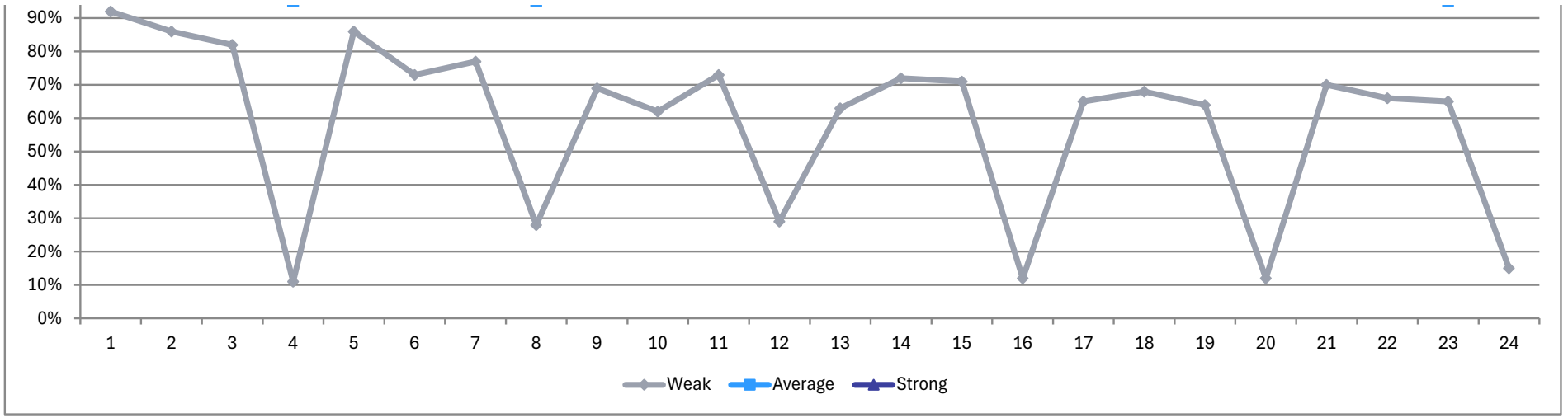
The climbs

300 climbs per player per level, each starting from one of that player's charges. Paint is per win.

#	Level	Chapter	Hard?	Guardian	Smallest winning board	Weak win	Weak paint	Weak stars	Average win	Average paint	Average stars	Strong win	Strong paint	Strong stars
1	First Steps	1		190	31	92%	64	1.1	100%	139	2.5	100%	168	2.7
2	Two Paths	1		220	33	86%	74	1.1	100%	181	2.5	100%	217	2.7
3	Mind the Gap	1		220	33	82%	73	1.1	97%	171	2.6	99%	204	2.8
4	The Shielded Way	1	Hard	380	57	11%	146	2.0	95%	214	2.6	100%	253	2.8
5	Gate Garden	1		410	37	86%	105	1.5	100%	212	2.5	100%	256	2.7
6	Crowded Lane	1		280	39	73%	105	1.4	96%	239	2.5	99%	290	2.8
7	Long Way Round	1		370	41	77%	112	1.2	99%	265	2.9	100%	302	3.0
8	The Narrow Climb	1	Hard	480	57	28%	170	2.0	95%	294	2.6	100%	351	2.7
9	Storm Front	2		630	43	69%	143	1.5	99%	303	2.5	100%	373	2.7
10	The Gallery Gate	2		540	45	62%	142	1.1	100%	359	2.8	100%	415	3.0
11	Cypress Row	2		500	45	73%	131	1.3	98%	292	2.5	100%	351	2.7
12	Golden Rise	2	Hard	2250	45	29%	259	2.0	97%	417	2.6	100%	506	2.7
13	Swirling Sky	2		590	45	63%	156	1.1	98%	355	2.8	100%	408	2.9
14	The Reapers' Path	2		620	45	72%	152	1.2	99%	350	2.5	100%	413	2.7
15	Hay Bales	2		730	45	71%	168	1.5	97%	367	2.5	100%	455	2.7
16	Windbreak	2	Hard	1050	57	12%	328	2.4	97%	394	2.6	99%	469	2.8
17	Kitchen Steps	3		450	45	65%	159	1.5	98%	335	2.5	99%	409	2.8
18	The Bread Basket	3		770	45	68%	174	1.5	98%	367	2.5	100%	447	2.7
19	Blue Apron	3		650	45	64%	166	1.1	99%	424	2.8	100%	484	3.0
20	Window Light	3	Hard	2950	57	12%	376	2.4	96%	443	2.6	100%	515	2.7
21	Pouring Milk	3		510	45	70%	165	1.3	99%	396	2.5	99%	486	2.7
22	The Foot Warmer	3		790	45	66%	196	1.1	100%	513	2.9	100%	583	3.0
23	Old Delft	3		930	45	65%	205	1.3	95%	453	2.5	100%	536	2.7
24	The Masterpiece	3	Hard	1400	57	15%	427	1.6	97%	620	2.9	99%	698	3.0

Guardians are set from how a careful weak player actually walks the level, not the perfect route (which can reach a far x2 gate no careful player takes). Every chapter's 4th and 8th climb is a hard level sized for an average board: a weak player wins it 11-29% of the time and replays it, an average player 95% or more. Stars: 2 for 85% of the paint a 90 board can bank, 3 for 90% of what a 140 board can; an average player averages about 2.5, a weak one about 1.3. Paint per enemy is round(sqrt(power)), so income grows gently with the monsters.




90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
1
2
3
4
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9
10
11
12
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20
21
22
23
24
Weak
Average
Strong

The restorations

The bot restores each quarter with plenty of paint. Change a chapter's bucket price (blue) and the paint updates.

67	
1 The Great Wave (Hokusai)	6
2 Wheat Field with Cypresses (Van Gogh)	10
3 The Milkmaid (Vermeer)	15

Per quarter

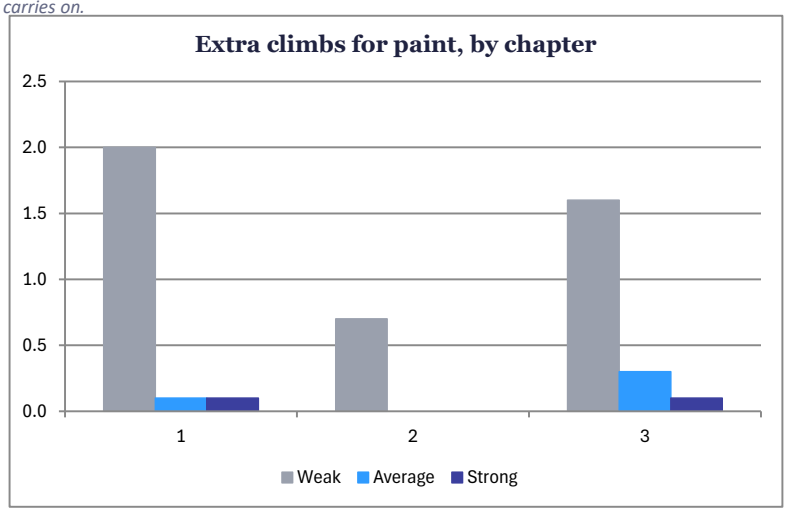
Painting	Quarter	Chapter	Buckets placed	Buckets paid (net of refunds)	Paint	Red	Yellow	Blue	Green	Purple	Sand: red	Yellow	Blue	Green	Purple	Dust
great-wave	1	1	19	17	104	0	52	52	0	0	0%	25%	22%	0%	0%	53%
great-wave	2	1	16	16	96	24	72	0	0	0	0%	41%	0%	0%	0%	59%
great-wave	3	1	16	16	96	0	0	96	0	0	0%	1%	33%	0%	0%	66%
great-wave	4	1	19	17	100	0	25	75	0	0	0%	5%	20%	0%	0%	74%
wheat-field	1	2	28	18	182	8	20	144	10	0	0%	3%	34%	4%	0%	59%
wheat-field	2	2	51	20	196	0	6	108	82	0	0%	3%	27%	21%	0%	50%
wheat-field	3	2	65	21	208	0	127	9	72	0	0%	53%	3%	27%	0%	17%
wheat-field	4	2	89	22	224	0	121	13	90	0	0%	47%	7%	30%	0%	16%
milkmaid	1	3	19	16	238	0	235	2	1	0	0%	85%	1%	1%	0%	13%
milkmaid	2	3	16	16	240	45	195	0	0	0	0%	24%	0%	0%	0%	76%
milkmaid	3	3	52	20	293	85	109	81	18	0	8%	28%	12%	0%	0%	51%
milkmaid	4	3	32	18	266	71	138	57	0	0	16%	25%	8%	0%	0%	51%

A quarter is 40 x 54 grains; whites, greys and blacks are dust that pours into any bucket. 80% drained and the brush sweeps the rest. A bucket holds 120 grains; one sent away early refunds its unfilled share, so 'buckets paid' is below 'buckets placed' where the bot swaps colours often (the Wheat Field, whose yellows and greens are woven together, is the busiest). The price per bucket rises by chapter (6 / 10 / 15) because the climbs pay more as the monsters grow.

The economy

40 whole journeys per player: each chapter's 8 climbs (a lost climb is replayed), then its 4 restorations; short of paint, the player replays the last climb and carries on.

Player	Chapter	Climbs played	Extra climbs for paint	Restoration visits	Paint earned	Paint spent	Paint left
weak	1	22.1	2.0	6.0	997	388	609
weak	2	21.7	0.7	4.7	1,591	802	789
weak	3	27.3	1.6	5.6	2,184	1,040	1,144
average	1	8.3	0.1	4.1	1,698	397	1,301
average	2	8.2	0.0	4.0	2,732	810	1,922
average	3	8.5	0.3	4.3	3,605	1,042	2,563
strong	1	8.1	0.1	4.1	2,030	397	1,633
strong	2	8.0	0.0	4.0	3,412	810	2,602
strong	3	8.1	0.1	4.1	4,087	1,039	3,048



Targets

Player	Max extra climbs a chapter	Worst chapter	On target?
weak	**2.5**	2.0	Yes
average	**0.5**	0.3	Yes
strong	**0.5**	0.1	Yes

Before phase 5 a bucket cost 5 in every chapter and the climbs paid 5-10 times what the restorations needed, while a weak player was still 9 climbs short in chapter 2: the paint came in the wrong colours. Three changes fixed it: each chapter's climbs now drop the colours its painting needs (Colours sheet), the bucket price rises by chapter, and a restoration that runs out of paint keeps the sand already drained, so the player climbs once more and carries on instead of paying twice. The paint left over (wrong colours, mostly) carries into the next chapter and is the stock for the boosters and shop of phase 6.

Paint by colour

What a chapter's 8 climbs pay in each colour, against what its painting needs at the bucket prices on the Restorations sheet.

Chapter 1: The Great Wave (Hokusai)

	Red	Yellow	Blue	Green	Purple
Painting needs	24	149	223	0	0
weak climbs pay	80	225	393	74	67
weak coverage	3.3x	1.5x	1.8x		
average climbs pay	160	429	842	159	87
average coverage	6.7x	2.9x	3.8x		

Chapter 2: Wheat Field with Cypresses (Van Gogh)

	Red	Yellow	Blue	Green	Purple
Painting needs	8	274	274	254	0
weak climbs pay	107	342	494	406	130
weak coverage	13.4x	1.2x	1.8x	1.6x	
average climbs pay	118	707	925	831	151
average coverage	14.8x	2.6x	3.4x	3.3x	

Chapter 3: The Milkmaid (Vermeer)

	Red	Yellow	Blue	Green	Purple
Painting needs	201	677	140	19	0
weak climbs pay	461	879	360	120	96
weak coverage	2.3x	1.3x	2.6x	6.3x	
average climbs pay	920	1584	709	181	96
average coverage	4.6x	2.3x	5.1x	9.5x	

Coverage under 1.0x (red) means the base climbs alone don't pay for that colour; the player makes it up with an extra climb or by mixing (green = yellow + blue, purple = red + blue, at 4/5 of a bucket of each). Each chapter's colour cycle leans on its painting: the Great Wave is blue then yellow, the Wheat Field an even yellow, blue and green, the Milkmaid mostly yellow with red and blue. Over a whole chapter even a weak player is covered; the one or two extra climbs on the Economy sheet come from timing. A restoration arrives after every two climbs, and a quarter that is nearly all one colour (the Great Wave's third, all blue) can ask for more of it than the last two climbs paid. The saved sand makes that a short detour rather than a wall.

The meta

The daily painting, the weekend Grand Climb and the skins shop (phase 6).

Daily painting: the box of buckets per quarter (the bot's need x1.3, plus one spare of every paint used)

Painting	Quarter	Red	Yellow	Blue	Green	Purple	Buckets	Bot result	Buckets left
great-wave	1	0	13	13	0	0	26	Restored	8
great-wave	2	7	17	0	0	0	24	Restored	8
great-wave	3	0	0	22	0	0	22	Restored	6
great-wave	4	0	7	18	0	0	25	Restored	8
wheat-field	1	3	4	20	3	0	30	Restored	11
wheat-field	2	0	2	16	12	0	30	Restored	10
wheat-field	3	0	18	3	11	0	32	Restored	11
wheat-field	4	0	17	3	13	0	33	Restored	11
milkmaid	1	0	22	2	2	0	26	Restored	10
milkmaid	2	5	18	0	0	0	23	Restored	7
milkmaid	3	9	11	9	3	0	32	Restored	12
milkmaid	4	8	13	6	0	0	27	Restored	9

Grand Climb: best tier of three tries, over twelve generated weekends

Player	No tier	Bronze	Silver	Gold
weak	8%	90%	3%	0%
average	0%	2%	55%	44%
strong	0%	0%	29%	71%

Weekend	Bronze line	Silver line	Gold line	Weak bronze+	Average silver+	Strong gold
17/10/2026	860	11,500	21,600	98%	100%	32%
24/10/2026	860	11,450	21,450	91%	101%	88%
31/10/2026	830	11,000	20,600	90%	100%	96%
07/11/2026	1,000	13,350	25,050	94%	101%	74%
14/11/2026	790	10,500	19,700	93%	101%	77%
21/11/2026	860	11,400	21,400	98%	99%	9%

28/11/2026	970	12,900	24,200	89%	99%	76%
05/12/2026	840	11,200	21,000	83%	97%	75%
12/12/2026	910	12,100	22,650	97%	101%	78%
19/12/2026	780	10,450	19,600	91%	87%	82%
26/12/2026	710	9,500	17,800	98%	100%	100%
02/01/2027	820	10,900	20,400	88%	100%	68%

Skins: the paint sink (prices are inputs; leftover paint per chapter from the Economy sheet)

Average player's paint left a ch 1,929

Weak player's paint left a chap 847

Skin	Price	Unlock	Running total	Chapters (average)	Chapters (weak)
Whiskers (Cat)	**0**	Default	0	0.0	0.0
Biscuit (Dog)	**300**	Paint	300	0.2	0.4
Clucky (Chicken)	**600**	Paint	900	0.5	1.1
Coo (Pigeon)	**900**	Paint	1,800	0.9	2.1
Sunny (Birb)	**1,300**	Paint	3,100	1.6	3.7
Merlin (Wizard)	**1,800**	Paint	4,900	2.5	5.8
Shadow (Ninja)	**2,500**	Paint, or the start	7,400	3.8	8.7
Zib (Alien)	**0**	Grand Climb gold	7,400	3.8	8.7

The daily box leaves the bot about a third of its buckets, room for a person's slower play. The Grand Climb is drawn so that every row can be survived from every lane by a careful hero (a safe middle, or safe edges); feasts beside the safe cells are what a bigger board eats, so the board decides how high you go. The lines are 0.6x, 8x and 15x the power a careful 100-power hero reaches. Skins are paid from the biggest piles first, so they soak up the colours the paintings don't need: over the three chapters an average player's leftover paint buys all but the last skin (the daily and Grand Climb rewards cover the rest), and a weak player's buys two or three.

Tuning

Every tuning number, as the build uses it (ChargeRules, ClimbRules, RestorationRules).

Area	Field	Value	What it does
Charge	**Seconds**	30	Length of the match-3 charge
Charge	**StartPower**	20	Hero power before the first move
Charge	**PowerPerTile**	0.5	Power per tile cleared
Charge	**PowerPerCascade**	0	Bonus per cascade after the first
Charge	**PowerPerSpecial**	3	Bonus per special tile made
Charge	**ComboMultiplier**	1.4	Power multiplier when two specials are swapped together
Charge	**MaxShields**	2	Shields carried into the climb (from Star specials)
Climb	**PaintFor(power)**	round(sqrt(power))	Paint dropped by an enemy of that power
Climb	**MagnetRatio**	0.5	Paint pulled from a neighbouring lane with the magnet
Climb	**GuardianJackpot**	2	Paint multiplier for beating the guardian
Climb	**Stars**	0.85 / 0.90	Shares of the solver's best paint (boards 90 / 140) for 2 / 3 stars
Restoration	**Width x Height**	40 x 54	Grains in a quarter of a painting
Restoration	**Funnels**	4	Funnels under the painting
Restoration	**BucketCapacity**	120	Grains a bucket holds
Restoration	**ChapterBucketCost**	10/06/2015	Paint per bucket in chapters 1 / 2 / 3
Restoration	**MixCostEach**	4/5 of a bucket	Paint of each parent colour to mix a green or purple
Restoration	**WinShare**	0.8	Share of the sand drained before the brush sweeps the rest
Restoration	**Refund**	unused share	A bucket sent away early refunds its unfilled share of the price
Restoration	**Saved sand**	kept	Running out of paint keeps the drained sand for the next visit
Daily	**Box**	bot x1.3 + 1	Buckets in the daily's box: what the bot spends, plus a margin and a spare of each paint
Daily	**Reward**	(2 + streak) buckets	Paint for the first win of the day, at the current chapter's bucket price; the streak counts up to 7
Daily	**UnlockStep**	3	The daily and the Grand Climb open after the first restoration
Grand	**Rows**	20	Rows in a weekend's Grand Climb
Grand	**Lines**	0.6 / 8 / 15	Bronze / silver / gold, as multiples of a careful 100-power hero's power at the top
Grand	**Tries**	3 + 2	Free tries a weekend, plus tries for rewarded ads
Grand	**Rewards**	150 / 300 / 600	Paint per tier, once a weekend each; first gold ever gives the Zib skin
Ads	**RewardedPerDay**	12	Cap on rewarded ads a day
Ads	**Interstitials**	from level 8, every 3 climbs	At least 180 s apart, 90 s after a rewarded ad; never with the starter pack or ad removal
Ads	**RefillBuckets**	3	Buckets a restoration or daily refill gives